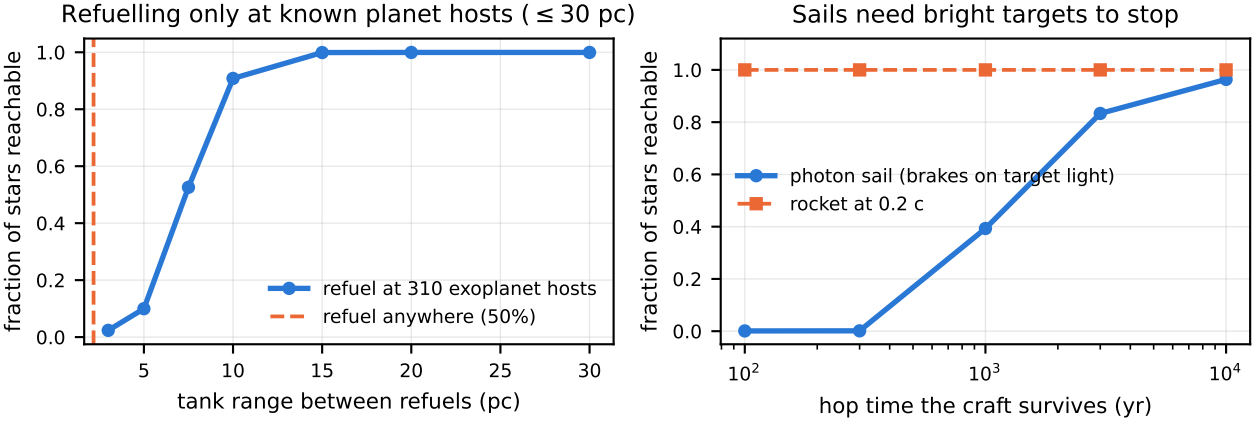
Refuelling only at known planet hosts (≤ 30 pc)
fraction of stars reachable
1.0
0.8
0.6
0.4
0.2
0.0
5
10
15
20
25
30
tank range between refuels (pc)
refuel at 310 exoplanet hosts
refuel anywhere (50%)
Sails need bright targets to stop
fraction of stars reachable
photon sail (brakes on target light)
rocket at 0.2 c
10^2
10^3
10^4
hop time the craft survives (yr)